

Data-to-LLM Architecture & RAG-Enabled Chatbot

Flow: User -> Chat UI -> Backend API -> Retriever -> LLM -> Response

Data Sources: Gold Layer Parquet + Documentation/FAQs Vector Store

Application Layer

- Chat Interface
- Backend API orchestration
- RAG Retriever
- LLM response generation

Infrastructure Layer

- Gold data storage
- Vector DB (FAISS/Pinecone)
- FastAPI backend
- LLM endpoint

RAG Interaction

- Query embedding
- Retrieve structured + unstructured context
- Prompt augmentation
- Grounded answer generation